Literature Review

Building an Integrated Healthcare Data Platform: From Operational Hospital Data to Management Analytics

1. Introduction

1.1 Definition of the topic

This review examines the published literature on how hospitals can turn their day-to-day operational data into reliable information for management decisions. The project under consideration proposes an **integrated healthcare data platform** with two main parts. The first is an **operational database** that records daily hospital activity such as patients, admissions, diagnoses, staff, procedures, insurance claims, surgical supplies and safety incidents. The second is an **analytical data warehouse** that reorganizes this data around key business processes so that managers can analyse it easily.

Between the two, a **layered data pipeline** collects the raw data, checks and cleans it, and prepares it for analysis. The final output is a set of **management dashboards** showing measures such as admissions, length of stay, readmissions, costs, staff workload, claim outcomes and supply availability. The thesis of this review is that this layered approach, grounded in established data management research, is an appropriate and justified way to give hospital managers a single, trustworthy view of performance.

1.2 Scope of the review

The review covers five areas of literature that match the main stages of the project: (1) the role of data and analytics in healthcare management; (2) the difference between operational systems and analytical systems; (3) data warehousing and dimensional modelling; (4) data integration and layered data pipelines; and (5) data quality, synthetic data and good practice for reliable, repeatable data work. Peer-reviewed journal and conference papers, foundational textbooks and a small number of authoritative industry sources published in English are included. Older foundational works are kept because the core ideas of data management have remained stable.

1.3 Intentional exclusions

This review will **not** explore clinical decision support, machine learning or prediction models, real-time data streaming, medical data exchange standards, or comparisons between specific software products and vendors. It also does not cover detailed database design rules or software installation and configuration. Legal and privacy compliance is mentioned only to explain why synthetic data is used. These topics are relevant to healthcare data, but they fall outside the management analytics focus of this project.

1.4 General findings and availability of sources

Most sources agree on four points. First, hospitals generate large amounts of data that remain under-used for management because it is stored in systems designed to record care, not to analyse it. Second, operational and analytical work should be kept in separate systems. Third,

organizing data around business processes (dimensional modelling) remains the most widely used and understandable structure for reporting. Fourth, poor data quality is the greatest practical risk when hospital data is reused for analysis.

Sources on data warehousing and healthcare analytics are plentiful and well established. Sources on modern layered pipelines are newer, and many come from industry rather than academic research. Published examples that combine all stages, from operational data to management dashboards, in one documented healthcare project are scarce, which is the gap this project addresses.

2. Main Body

The sources are organized **thematically**, following the journey of data through the proposed platform: from the healthcare problem, to the operational source, through integration and quality checks, into the warehouse and finally to the dashboards. Each work is summarized and evaluated for its premise, methodology and conclusion, together with its relevance to the project.

2.1 Theme 1: Data and analytics in healthcare management

Raghupathi and Raghupathi (2014) start from the **premise** that analysing large volumes of healthcare data can improve outcomes and reduce costs. Their **method** is a conceptual review that proposes a framework of data sources, data preparation, storage and analytical applications. They **conclude** that the potential is high but depends on integrating data, governing it well and having skilled staff. The framework is useful because its stages mirror this project's stages. However, the paper offers no practical evaluation, so it explains why analytics matters rather than how to achieve it.

Wang, Kung and Byrd (2018) move from promise to evidence. Their **premise** is that hospitals must understand which analytics capabilities produce benefits. Using a **multiple case study** of 26 healthcare organizations, they link capabilities such as analysing patterns of care and supporting decisions to operational, managerial and strategic benefits. Their **conclusion** that bringing different data sources together is a precondition for benefit directly supports building a central warehouse. A limitation is that the cases are partly based on vendor descriptions, which may overstate success.

Together, these works establish the **context**: hospital managers need integrated, analysis-ready data on admissions, length of stay, readmissions, staffing, claims and supplies. They do not explain how such data should be organized, which leads to the next theme.

2.2 Theme 2: Operational systems versus analytical systems

Codd (1970) laid the foundation for modern operational databases. His **premise** is that data should be stored in a structured, logical way that is independent of how computers physically hold it. Through a formal **theoretical method**, he shows that well-organized data avoids duplication and inconsistent updates. His **conclusion** supports keeping the hospital's existing operational database design, because it protects the accuracy of patient, admission and claim records as they are created and updated. However, such designs are built for recording transactions, not for reporting, and analysing them directly can be slow and complex.

Chaudhuri and Dayal (1997) address exactly this problem. Their **premise** is that decision support needs a different kind of system from daily transaction processing. In a structured **survey** of data warehousing and analytical processing, they **conclude** that analysis should run on a separate store

that is integrated, historical and organized for querying. This is the clearest justification in the literature for the project's two-part design of an operational database and a separate analytical warehouse. The paper is dated in its technology, but its central principle remains widely accepted.

2.3 Theme 3: Data warehousing and dimensional modelling

Two main schools of thought shape warehouse design. Inmon (2005) defines a data warehouse as a subject-oriented, integrated, time-variant and non-volatile collection of data, and recommends first building one enterprise-wide integrated store. His approach, based on extensive practice, favours long-term consistency but can be slow to deliver visible results.

Kimball and Ross (2013) take a more incremental path. Their **premise** is that warehouses should be built one business process at a time, with each process represented by **facts** (the numbers being measured, such as length of stay or claim value) and **dimensions** (the context for analysis, such as patient, date, hospital unit or diagnosis). Shared dimensions allow results from different processes to be compared. Their **method** is a practical design process illustrated with case studies, including healthcare. They **conclude** that this structure is easy for business users to understand and fast to analyse, and that history should be preserved when details change, for example when a patient changes insurer.

This project follows the Kimball approach because it supports delivering one complete process first (patient admissions) and then adding others such as procedures, claims, staffing and supplies. The two schools are not fully opposed: the project's cleaned, integrated middle layer resembles Inmon's idea of an integrated store, making the overall design a practical hybrid. A weakness of this literature is that it relies mainly on practitioner experience rather than formal evaluation.

Healthcare-specific research confirms that these ideas work in hospitals. Murphy et al. (2010) describe **i2b2**, a clinical research warehouse adopted by many institutions, and **conclude** from its deployment that a simple, shared structure can serve many users. Hripcsak et al. (2015) describe the **OHDSI** community and its common data model, showing that standardizing data enables analysis across many organizations. Both, however, focus on clinical research, such as studying patient groups, rather than hospital management measures such as nurse workload, claim denials or supply availability. This difference is central to the project's contribution.

2.4 Theme 4: Data integration and layered data pipelines

Vassiliadis (2009) reviews the process of extracting, transforming and loading data into warehouses. His **premise** is that this integration step is the most time-consuming and error-prone part of warehousing. Through a **literature survey**, he **concludes** that integration should be carefully designed and tested rather than handled through informal scripts. This supports the project's decision to treat data preparation as a series of clear, testable steps.

More recently, Armbrust et al. (2021) argue that modern data platforms should keep data in open, low-cost formats and refine it in stages. Their **position paper**, supported by performance results, **concludes** that this reduces cost and complexity compared with older designs. Industry sources describe this staged refinement as a **layered (raw, cleaned, business-ready) architecture** (Databricks, n.d.). Each layer has a clear purpose: the raw layer keeps an untouched copy of the source data, the cleaned layer fixes errors and standardizes values, and the business-ready layer is organized for analysis. The project adopts this layered approach because it makes every step traceable and easy to check. A limitation is that the layered approach is described mainly by industry rather than tested in peer-reviewed studies, which is itself a gap noted in this review.

2.5 Theme 5: Data quality in healthcare

Wang and Strong (1996) define data quality from the user's point of view. Using a **two-stage survey** of data users, they identify groups of quality dimensions and **conclude** that quality means "fitness for use", not only accuracy. Batini et al. (2009) compare 13 methods for assessing data quality and **conclude** that assessment should be systematic and cover completeness, accuracy, consistency and timeliness.

Kahn et al. (2016) apply these ideas to electronic health records. Through an **expert consensus process**, they propose three practical categories of checks: **conformance** (values follow the expected format and codes), **completeness** (required data is present) and **plausibility** (values make sense, for example a discharge date after the admission date and no negative lengths of stay). This framework is the most directly usable source for the project, because it turns into concrete checks in the cleaned layer. A limitation shared by these works is that they define what to check but give little guidance on building checks into routine data processing.

2.6 Theme 6: Synthetic healthcare data

Real patient data cannot be used freely because of privacy rules. Walonoski et al. (2018) present **Synthea**, an open tool that generates realistic but artificial patient records based on public health statistics. They **conclude** that such data can safely support development and teaching. El Emam, Mosquera and Hoptroff (2020) review synthetic data more broadly and note that it can reproduce the main patterns of real data while greatly reducing privacy risk. Both caution that synthetic data may miss rare events and real-world errors. For this project, the synthetic data should therefore deliberately include some problems, such as duplicates, missing values and impossible dates, so that the cleaning and quality checks are properly tested.

2.7 Theme 7: Reliable and repeatable data work

Good practice literature stresses that data projects should be repeatable, documented and reusable. The FAIR principles (Wilkinson et al., 2016) state that data and research outputs should be findable, accessible, interoperable and reusable. For this project, this means keeping all steps documented and version-controlled, running the pipeline automatically on a schedule, and recording errors so they can be investigated. The literature on these principles is strong in general terms but offers limited evaluated guidance for small healthcare analytics projects.

3. Synthesis: Purposes of this Literature Review

This section shows how the review fulfils each of the five purposes of a literature review.

3.1 Establishing context and foundational knowledge

The review places the project within established knowledge: the value of analytics in healthcare (Raghupathi & Raghupathi, 2014; Wang et al., 2018); the separation of operational and analytical systems (Codd, 1970; Chaudhuri & Dayal, 1997); the main approaches to warehouse design (Inmon, 2005; Kimball & Ross, 2013); existing healthcare warehouses (Murphy et al., 2010; Hripcsak et al., 2015); and the move towards layered data pipelines (Armbrust et al., 2021).

3.2 Identifying the research gap

Four gaps emerge from the literature:

- **Management focus is missing.** Well-known healthcare warehouses are designed for clinical research, not for hospital management measures such as staff workload, claim denials, safety incidents and supply availability.
- **Limited academic evidence on layered pipelines.** The raw, cleaned and business-ready approach is widely used but mainly described by industry.
- **Few complete, documented examples.** Most sources address one stage, such as design, integration or quality, in isolation. Documented projects that cover the full journey from operational data to dashboards are scarce.
- **Quality checks are rarely built in.** Quality frameworks describe what to check but rarely show how to make checking a routine part of data processing.

3.3 Justifying the research and methodology

Each main design decision in the project is supported by the literature, as summarized below.

Project decision	Justification from literature
Keep the existing operational database design	Well-structured operational data protects accuracy (Codd, 1970).
Build a separate analytical warehouse	Analysis should be separated from daily operations (Chaudhuri & Dayal, 1997).
Organize data into facts and dimensions	Dimensional modelling is understandable and efficient for reporting (Kimball & Ross, 2013).
Start with patient admissions, then expand	Incremental, process-by-process delivery (Kimball & Ross, 2013).
Raw, cleaned and business-ready layers	Staged refinement improves traceability and control (Armbrust et al., 2021; Databricks, n.d.).
Designed and tested data preparation	Integration is the most error-prone stage (Vassiliadis, 2009).
Conformance, completeness and plausibility checks	Healthcare data quality framework (Kahn et al., 2016; Wang & Strong, 1996).
Use synthetic patient data	Realistic data without privacy risk (Walonoski et al., 2018; El Emam et al., 2020).
Documented, automated and reusable work	FAIR principles (Wilkinson et al., 2016).

3.4 Avoiding duplication of effort

The review shows that the project does not need to invent new theory. It reuses proven ideas, such as separating operational and analytical data, dimensional modelling, staged data refinement and established quality checks. It also avoids repeating existing clinical research warehouses, which already serve that purpose well, and instead focuses on the under-served area of hospital management analytics. Where existing solutions exist, such as tools for generating synthetic patients, the project adopts them rather than building its own.

3.5 Formulating the conceptual framework

The literature provides the measures, structures and checks for the project. The framework below links each business area to what is measured and the sources that support it.

Business area	What is measured	Main sources
Patient admissions	Number of admissions, length of stay, readmissions, cost per stay	Kimball & Ross (2013); Wang et al. (2018)
Procedures	Procedure volume and duration	Kimball & Ross (2013)
Nursing staff	Hours worked, patients per nurse	Raghupathi & Raghupathi (2014)
Equipment	Usage and utilization rate	Kimball & Ross (2013)
Insurance claims	Claim value, amount paid, denial rate	Wang et al. (2018)
Patient satisfaction	Satisfaction scores	Wang et al. (2018)
Patient safety	Number and severity of incidents	Raghupathi & Raghupathi (2014)
Surgical supplies	Stock levels and availability over time	Kimball & Ross (2013)

The framework is completed by three cross-cutting ideas from the literature: **shared dimensions** such as patient, date, hospital unit and diagnosis, used across all business areas (Kimball & Ross, 2013); **quality checks** for conformance, completeness and plausibility (Kahn et al., 2016); and **staged refinement** of data from raw to business-ready (Armbrust et al., 2021).

4. Conclusion

The literature consistently shows that hospitals hold valuable data that is hard to analyse in their operational systems, that analytical work should be kept separate from operations, that organizing data by business process is the most effective structure for reporting, and that data quality must be managed deliberately. Newer work on layered pipelines adds a staged, traceable way of preparing data, although it is supported mainly by industry rather than academic evidence.

A common feature of the sources is that each covers only one part of the data journey. Existing healthcare warehouses serve clinical research rather than management, and quality frameworks are rarely built into routine processing. These gaps justify the proposed project: **an integrated, documented healthcare data platform** that keeps the hospital's operational database as the source, prepares data through raw, cleaned and business-ready layers with built-in quality checks, and loads an analytical warehouse that powers management dashboards. Beginning with patient admissions and expanding to procedures, claims, staffing and supplies, the project applies proven ideas to an under-served area and offers a reusable example for hospital management analytics.

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